Xiongxin Yang

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Education

University of California, Santa Barbara — Sept 2025 – Present
Ph.D. Student in Computer Science
- Advisor: Eric Vigoda

Northeast Normal University — Sept 2020 – Jun 2024
B.S. in Computer Science and Technology
- Advisor: Zhiguo Fu

Employment

The University of Hong Kong — May 2025 – Jul 2025
Research Assistant hosted by Weiming Feng

New York University Shanghai — Oct 2024 – Jan 2025
Research Associate hosted by Jie Xue

Visits & Internships

The University of Edinburgh — Jul 2026
Visiting Student hosted by Heng Guo

The University of Hong Kong — Mar 2025
Visitor hosted by Weiming Feng

Nanjing University — Apr 2023
Visiting Student hosted by Yitong Yin

Selected Awards

Presidential Scholarship (Northeast Normal University) — 2022 & 2024
Ranked 1st in the major

National Scholarship — 2021
Top national scholarship in mainland China (top 0.2% nationwide)

Publications & Preprints

[10] An $\widetilde{O}(n^3)$-time FPRAS for the Permanent
Xiaoyu Chen, Heng Guo, Eric Vigoda, **X. Yang**
arXiv: 2609.20717

[9] Optimal Simulated Annealing for Partition Function Estimation
Heng Guo, Hongyang Liu, **X. Yang**, Yitong Yin, Yiyao Zhang
arXiv: 2609.20337

[8] Faster FPRAS for the Permanent via Restricted Poincaré Inequalities and Coupled Flows
Xiaoyu Chen, Eric Vigoda, **X. Yang**
arXiv: 2608.26599

[7] 0/1-Polytopes with Exponentially Small Edge Expansion
X. Yang
arXiv: 2608.01870

[6] Learning AC^0 under Locally Sampleable Graphical Models
Weiming Feng, **X. Yang**, Yixiao Yu, Yiyao Zhang
2027 ACM-SIAM Symposium on Discrete Algorithms (**SODA 2027**)

[5] Near-Optimal Dynamic Data Structures for Maximum Depth and Klee's Measure of Boxes
Sujoy Bhore, Subhash Suri, Jie Xue, **X. Yang**, Jiumu Zhu
53rd EATCS International Colloquium on Automata, Languages, and Programming (**ICALP 2026**)

[4] Learning CNF formulas from uniform random solutions in the local lemma regime
Weiming Feng, **X. Yang**, Yixiao Yu, Yiyao Zhang
58th Annual ACM Symposium on Theory of Computing (**STOC 2026**)

[3] Dynamic maximum depth of geometric objects
Subhash Suri, Jie Xue, **X. Yang**, Jiumu Zhu
41st International Symposium on Computational Geometry (**SoCG 2025**)

[2] Spectral Independence Beyond Total Influence on Trees and Related Graphs
Xiaoyu Chen, **X. Yang**, Yitong Yin, Xinyuan Zhang
2025 ACM-SIAM Symposium on Discrete Algorithms (**SODA 2025**)

[1] Beyond windability: Approximability of the four-vertex model
Tianyu Liu, **X. Yang**
Theoretical Computer Science, Volume 995 (2024), Article 114491

Talks & Posters

- Near-Optimal Dynamic Maximum Depth and Klee's Measure of Boxes
  - Jul 8, 2026 @ ICALP 2026: Royal Holloway, University of London, UK
- Learning CNF formulas from uniform random solutions in the local lemma regime
  - Jun 22, 2026 @ STOC 2026 Online Poster Session
- Dynamic Maximum Depth of Geometric Objects
  - Jun 25, 2025 @ SoCG 2025: Kanazawa, Ishikawa Prefecture, Japan
  - May 31, 2025 @ AAAC 2025: The City University of Hong Kong, Hong Kong SAR, China
- A Glimpse into the Theory World
  - Apr 11, 2025 @ Information Science Academic Forum: Northeast Normal University, Changchun, China
- Approximability of the Four-Vertex Model
  - Oct 21, 2024 @ NYUSH Theory Lunch: New York University Shanghai, Shanghai, China

Services

Conference Reviewing: WWW (2026), SoCG (2025), SODA (2027), SOSA (2027)